P-P-105b
September 10, 1964

SUPERSEDING
Int. Fed. Spec. P-P-00105a (GSA-FSS)
August 22, 1962 and
Fed. Spec. P-P-105
December 11, 1952

FEDERAL SPECIFICATION

PAPER, ABRASIVE (FLINT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Flint paper covered by this specification is a paper coated on one surface with a flint abrasive.

1.2 Classification.

1.2.1 *Classes and grit sizes.* Flint paper shall be of the following classes and grit sizes, as specified (see 6.2):

Class 1—Sheets, flint finishing paper (see 3.4.1).
Grit size—7/0, 6/0, 5/0, 4/0, 3/0, 2/0.

Class 2—Sheets, flint paper (see 3.4.1).
Grit size—Extra fine, fine, medium, coarse, extra coarse.

Class 3—Rolls, flint paper (see 3.4.2).
Grit size—Extra fine, fine, medium, coarse.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specification:

UU-P-31—Paper, General Specifications and Methods of Testing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-3816—Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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TABLE I—*Weight of paper backing*

Class No.	Grit size	Nominal paper weight, 24 by 36 inches, 480 sheets	Tolerance	Symbol or designation
		Pounds	*Pounds*	
1	7/0 through 2/0	40	37.5— 42.5	A
2	Extra fine, fine	50 to 60	46.5— 63.0	—
2	Medium, coarse, extra coarse	55 to 60	51.0— 63.0	—
3	Extra fine, through coarse	130	122.0—135.0	E

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publication.** The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Department Of Commerce

Commercial Standard:

CS217—Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Abrasive.*

3.1.1.1 *Flint.* The abrasive shall be crushed or ground from clean hard quartz, quartzite, or flint and shall be relatively free from extraneous materials.

3.1.1.2 *Grading, control, and tolerance.* Grading of abrasive particles shall be in accordance with CS217; and the tolerance therein shall apply to the abrasive grain recovered from the coated abrasive product.

3.1.2 *Backing.* The backing shall be paper and shall meet the requirements of tables I and II directly after being conditioned at 70° ± 2°F. and 50 ± 2 percent relative humidity for at least 24 hours.

3.1.3 *Adhesive.* The adhesive shall be a good quality, animal hide glue base, suitable for the purpose intended.

3.2 Coating.

3.2.1 *Class* 1. The flint finishing paper sheets shall be open coat. The abrasive flint grain shall be applied to one side of the paper backing only, and cover approximately 50 percent of the backing, thus leaving open spaces between the grains of abrasive (see 6.5).

3.2.2 *Classes* 2 *and* 3. The flint paper sheets and rolls shall be closed coat. The abrasive flint grain shall be applied to one side of the paper backing only, and shall completely cover the surface of the backing (see 6.5).

3.3 Tensile breaking strength. The average breaking strength of coated abrasive paper shall be in accordance with table II, when tested as described in 4.4.2.

TABLE II—*Tensile breaking strengths*

Class No.	Grit size	Tensile strength, average	
		Machine direction	Across machine direction
		Pounds/ Inch	*Pounds/ Inch*
1	7/0 through 2/0	20	10
2	Extra fine, fine	35	15
	Medium, coarse, extra coarse	40	20
3	Extra fine through coarse	85	45

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3.4 **Form and dimensions.**

3.4.1 *Sheet.* Unless otherwise specified (see 6.2), flint paper shall be furnished in the following sizes, with a tolerance of ± 1/8 inch in length and ± 1/16 inch in width, when measured at, and after conditioning at 70° ± 2°F. and 50 ± 2 percent relative humidity for at least 24 hours:

Class 1—9 by 11 inches
Class 2—9 by 10 inches

3.4.2 *Roll.* Class 3 flint paper shall be furnished in rolls, not less than 50 yards long, and in the widths as specified (see 6.2).

3.5 **Identification marking.** Each sheet of class 1 and class 2 abrasive paper and each roll of class 3 abrasive paper shall be marked in a legible and permanent manner with the following information. Class 3 rolls shall be marked at least once in every lineal yard.

(a) Manufacturer's name or trademark of such character that the source of manufacture may be readily determined.
(b) Class and grit size.

3.6 **Workmanship.** Abrasive flint paper shall be free from imperfections affecting its appearance or which may impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling.** The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 *Lot.* All abrasive paper of the same class and grit size manufactured under essentially the same conditions and offered for delivery at one time shall be considered a lot for the purposes of examinations and tests.

4.2.2 *Sampling for examination of packaging and packing.* Packaged and packed abrasive paper in shipping containers shall be sampled in accordance with inspection level I.

4.2.3 *Sampling for visual, dimensional, and workmanship characteristics of end item.* Sampling for these characteristics shall be in accordance with inspection level S-4.

4.2.4 *Sampling for tests.* Samples for test shall be selected in accordance with inspection level S-2. Samples may be selected from those selected in accordance with 4.2.3.

4.3 **Examination and tests.** The acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

4.3.1 *Examination.*

4.3.1.1 *Packaged and packed abrasive paper.* Abrasive paper, ready for shipment, selected in accordance with 4.2.2, shall be examined with AQL's and classification of defects in accordance with MIL-P-3816.

4.3.1.2 *Individual sheets and rolls.* Samples of abrasive paper selected in accordance with 4.2.3 shall be examined in accordance with table III. The AQL's for defects listed in table III shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE III—*Classification of defects, end item*

Examine (applicable paragraph)	Defects	Major	Minor
1.2.1	Size of grit not as specified	X	
3.1.1.1	Extraneous material present in abrasive	X	
3.2	Abrasive not closed or open coat as specified	X	
3.2	Coating not completely covering surface of backing	X	
3.4.1	Dimensions of sheets, not correct size or not within tolerances	X	
3.4.2	Size of rolls not as specified	X	
3.5	Identification marking, omitted, illegible, incomplete, or incorrect		X
3.6	Sheets and rolls, torn or cracked	X	

4.3.2 *Lot acceptance tests.* Samples selected in accordance with 4.2.4 shall be subjected to the tests in 4.4, AQL 10.0 percent defective.

4.4 **Tests**

4.4.1 *Grain size.*

4.4.1.1 *Grading for grain size* Grading to determine the grain size, including recovery of abrasive grain from the sheets, preparation of grain for grading, and procedures shall be in accordance with ~~CS217~~.

4.4.1.2 *Certification of grading.* Grading tests specified in 4.4.1.1 shall be omitted in instances where the contractor produces certification in writing that the abrasive grain complies with the requirements of ~~CS217~~ for flint. Contractors are urged to include this certification with material offered for inspection and inspectors are urged to determine whether such certification is included with such material in order to preclude the necessity for grading tests.

4.4.2 *Breaking strength.* The tensile breaking strength of the coated abrasive, directly after conditioning for at least 24 hours at 70° ± 2°F. and 50 ± 2 percent relative humidity shall be determined by method 171 of UU-P-31, except that the jaws of the testing machine shall be 5 inches apart, and that six specimens 1 inch wide shall be taken both longitudinally and transversely from sample sheets and rolls. The average tensile strength in both directions shall be computed from at least three specimens. Samples tearing at the jaws of the tensile testing machine should be discarded (see 3.3).

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing of abrasive paper shall be in accordance with MIL-P-3816, as specified for level A, B, or C (see 6.2).

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special markings required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to any special markings required in the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Abrasive flint paper is intended for all ordinary hand and machine sanding of wood or painted surfaces.

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6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following.

(*a*) Title, number, and date of this specification.

(*b*) Class and grit size for sheets (see 1.2).

(*c*) Class, grit size, and width of rolls (see 1.2 and 3.4.2).

(*d*) Level of packaging and level of packing required (see 5.1).

6.3 Flint. A white, translucent, somewhat crystalline, variety of natural silicon dioxide, having a structure such that when crushed develops sharp edges. The hardness is approximately 6.8-7.0 (Mohs scale), or 820 Knoop scale, with a specific gravity of approximately 2.6.

6.4 Coating. A closed coat is one in which the abrasive grain is put on in such quantity as entirely to cover the backing. An open coat covers about 50 percent of the backing thus leaving open spaces between the particles of abrasive grain.

6.5 Work value. Coated abrasives should be stored in a cool even temperature area where the relative humidity is somewhere between 40 to 50 percent. Do not store on concrete floors. The best work value will be obtained if stored and worked in rooms having a relative humidity of approximately 50 percent.

6.6 Abrasive grain size. The size of abrasive grain is indicated in table IV which shows the approximate relation in size between flint and artificial abrasives.

TABLE IV—*Comparative grit sizes*

Silicon carbide and aluminum oxide grit size	Class 1, flint	Classes 2 and 3, flint
320	7/0	
280	6/0	
240	5/0	
220	4/0	
180	3/0	Extra fine (EF)
150	2/0	
120		Fine (F)
100		
80		Medium (M)
60		
50		
40		Coarse (C)
36		
30		Extra coarse (EC)

6.7 Transportation description. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Abrasive paper.
Carload minimum weight 36,000 pounds.

Motor:

Abrasive paper.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MR

Navy—Ships

Air Force MOA

Copies of this specification may be purchased for ents each.

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